HSE Field Guide | SERNEM Pro Series | Site validation required

SERNEM Health and Safety Resources

HOT WORK CRITICAL SAFETY RULES

Authorize • Control Combustibles • Prevent Fire

OSHA 1910.252(a)(2)(iii)(A)(1–4), (B); (c)
OSHA · US general industry · Verify local requirements separately

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SERNEM · Check Before Starting

Pre-Job Check

- ✓ Permit where required
- ✓ Gas test where required
- ✓ Combustible/fume control
- ✓ Fire watch where required
- ✓ Suitable extinguisher
- ✓ Screens/barriers if needed
- ✓ Post-work watch if required

01 REQUIRED IF — Hot Work Permit

Applies when: A permit system requires it

- ✓ Verify a valid hot-work permit where the permit system requires one.
- ✓ Confirm the work area and scope match the authorization.
- ✓ Stop work and reassess the permit if conditions change.

02 REQUIRED — Prepare the Work Area

- ✓ Check combustibles; remove them or protect fixed items with fire-resistant covers.
- ✓ Check floors, openings and opposite surfaces for sparks/heat.
- ✓ Assess metal-fume exposure; provide process-matched ventilation/local exhaust.

03 REQUIRED IF — Atmospheric Testing

Applies when: Atmospheric hazards or permit conditions require testing

- ✓ Test the atmosphere before work where required.
- ✓ Check oxygen/flammable/toxic gases; a gas detector does not replace fume control.
- ✓ Use continuous or periodic monitoring when conditions may change.

04 REQUIRED IF — Fire Watch

Applies when: OSHA fire-watch criteria or site risk applies

- ✓ Where OSHA watch criteria or site risk applies, assign a trained fire watch with no conflicting duties.
- ✓ Ensure all exposed areas remain visible and accessible.
- ✓ Authorize the fire watch to stop work and raise the alarm.

05 REQUIRED — Firefighting Equipment

- ✓ Keep suitable extinguishing equipment immediately available.
- ✓ Verify equipment is inspected and ready for use.
- ✓ Confirm alarm and emergency communication arrangements.

06 REQUIRED IF — Control Sparks and Heat

Applies when: Sparks or heat can affect nearby people or materials

- ✓ Confine sparks, slag and hot fragments.
- ✓ Screen nearby workers from arc radiation and spatter.
- ✓ Assess heat conduction through walls, pipes and metal surfaces.

07 REQUIRED IF — Cylinder and Hose Safety

Applies when: Cylinders and gas hoses are used

- ✓ Keep cylinders upright, secured and away from heat.
- ✓ Inspect hoses, regulators and connections before use.
- ✓ Keep oxygen equipment free from oil and grease.

08 REQUIRED IF — Post-Work Fire Watch

Applies when: After work requiring a fire watch

- ✓ US general industry: at least 30 minutes after welding/cutting requiring a watch.
- ✓ Extend for stricter local rules, permits or risk; do not shorten the applicable OSHA minimum.
- ✓ Construction 1926.352(e): watch until no fire risk; no fixed 30 min. Check concealed/smouldering fires.

Stop Work Now

- **Stop work immediately if a required hot-work permit is missing**
- **Stop work if flammable or explosive atmospheric hazards are uncontrolled**
- **Stop work if combustibles are not removed or protected**
- **Stop work if a required fire watch is missing or suitable extinguishing equipment is unavailable**
- **Stop work if sparks and hot fragments cannot be contained**

Critical Controls

35 ft
Watch criterion: combustibles <10.7 m; beyond this distance: sparks, openings and conducted heat

≥30 MIN
General industry: after work requiring a watch; extend if stricter

PERMIT
Where required: verify scope/conditions

WATCH
Where required: trained and ready

Technical Reference · Validation Required
OSHA 1910.252(a)(2)(iii)(A)(1–4), (B); (c)
OSHA 1926.352(e)

Risk-Based PPE Selection

- **Face/Eye Protection** — Welding/cutting/grinding: suitable filter/shield
- **Safety Glasses** — Process-matched impact/eye protection
- **Heat-Resistant Gloves** — Heat and spatter
- **Safety Footwear** — Match heat, spatter and mechanical hazards
- **Flame-Resistant Clothing** — Process-matched flame protection

Select by risk/manufacturer; add respiratory/hearing PPE if needed. PPE does not replace ventilation.